

Accelerators and GPUs

ECM3446/ECMM461: High Performance Computing

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Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manager worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Syllabus Plan

- Motivation of and introduction to high-performance computing
- **Parallel computer architectures**: shared-memory and distributed-memory architectures, multi-core processors, **Graphics Processing Units (GPUs)**
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Lecture Units

- Unit 10.1: Background and history of accelerators
- Unit 10.2: GPUs in HPC
- Unit 10.3: NVIDIA Pascal architecture
- Unit 10.4: GPU programming for HPC

Workshops

For the workshop, we will revisit two different finite differencing methods of solving the diffusion equation.